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ELECTRIC MOTOR AND MOTOR-DRIVEN COMPRESSOR

Abstract

An electric motor includes processing circuitry. The processing circuitry is configured to superpose a harmonic on at least one of instruction values, and estimate the position of a rotor by a harmonic superposition method based on the harmonic superposed on the at least one of instruction values. The frequency of a harmonic includes a first frequency and a second frequency that is lower than the first frequency. The processing circuitry is further configured to determine whether a temperature-associated value becomes greater than a threshold value. When the temperature-associated value becomes greater than the threshold value, the processing circuitry is further configured to set the frequency of the harmonic for superposition on the at least one of instruction values to the second frequency.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-027574, filed on Feb. 27, 2024, and Japanese Patent Application No. 2024-228939, filed on Dec. 25, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

[0002] The following description relates to an electric motor and a motor-driven compressor.

2. Description of Related Art

[0003] When a motor is driven by sensorless control, the position of a rotor of the motor needs to be estimated by a controller. Sensorless control is a control method that drives a motor without using a hardware position sensor and estimates the position of a rotor using software.

[0004] A typical method for estimating the position of a rotor includes, for example, an induced voltage method and a harmonic superposition method. The induced voltage method estimates the position of a rotor based on an induced voltage generated by rotation of a motor. Japanese Laid-Open Patent Publication No. 2011-172324 describes a harmonic superposition method that estimates the position of a rotor based on a current (response current) flowing through a motor when a harmonic component is superposed on a voltage instruction value or a current instruction value to the motor.

SUMMARY

[0005] A motor loss may increase as the frequency of a carrier (carrier wave) for pulse width modulation (PWM) control of the motor and the frequency of a harmonic superposed for position estimation of a rotor become higher. In contrast, a decrease in the frequency of the harmonic may adversely affect the noise and vibration (NV) characteristics. Nonetheless, the above publication is mute on specific settings of the frequency of the harmonic.

[0006] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0007] In one general aspect, an electric motor includes a motor unit, an inverter, a temperature-associated value obtainer, and processing circuitry. The motor unit includes a rotor and a stator around which three-phase coils are wound. The inverter includes a driver and a switching element operated by the driver. The inverter is configured to drive the motor unit through operation of the switching element. The temperature-associated value obtainer is configured to obtain a temperature-associated value that is associated with a temperature of the motor unit. The processing circuitry is configured to calculate instruction values for control of the switching element and control the switching element. The processing circuitry is configured to superpose a harmonic on at least one of the instruction values. The processing circuitry is configured to estimate a position of the rotor by a harmonic superposition method based on the harmonic superposed on the at least one of instruction values. A frequency of the harmonic includes a first frequency and a second frequency that is lower than the first frequency. The processing circuitry is further configured to determine whether the temperature-associated value becomes greater than a threshold value. The processing circuitry is further configured to set the frequency of the harmonic for superposition on the at least one of the instruction values to the second frequency when the temperature-associated value becomes greater than the threshold value.

[0008] In another general aspect, a motor-driven compressor includes a compression unit configured to compress a fluid, and the above-described electric motor configured to drive the compression unit.

[0009] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a diagram showing an example of the configuration of a vehicle air conditioner.

[0011] FIG. 2 is a graph illustrating the relationship of a frequency of a harmonic switched by a controller of the vehicle air conditioner shown in FIG. 1 and an effective value of a phase current.

[0012] FIG. 3 is a flowchart illustrating an example of the series of processes executed by the controller of the vehicle air conditioner shown in FIG. 1 to control a motor unit.

[0013] FIG. 4 is a flowchart illustrating an example of the series of processes executed by the controller of the vehicle air conditioner shown in FIG. 1 to set the frequency of the harmonic.

[0014] FIG. 5 is a graph illustrating the relationship of a frequency of a harmonic switched by a controller of a modified example and an effective value of a phase current.

[0015] FIG. 6 is a flowchart illustrating an example of the series of processes executed by the controller of the modified example to set the frequency of the harmonic.

[0016] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0017] This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

[0018] Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

[0019] In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

Embodiment

[0020] An embodiment of an electric motor and a motor-driven compressor will now be described with reference to the drawings.

Overall Configuration

[0021] As shown in FIG. 1, a vehicle air conditioner **100** of the present embodiment includes a motor-driven compressor **101** and a refrigerant circuit **103**. The motor-driven compressor **101** includes a compression unit **102** and an electric motor **M1**. The motor-driven compressor **101** compresses refrigerant. The refrigerant circuit **103** includes, for example, a heat exchanger, an expansion valve, or the like. The motor-driven compressor **101** compresses refrigerant, and the refrigerant circuit **103** exchanges heat with the refrigerant and expands the refrigerant. This allows the vehicle air conditioner **100** to cool or warm the passenger compartment. The motor-driven compressor **101** discharges oil together with the compressed refrigerant.

[0022] The compression unit **102** compresses refrigerant. The refrigerant compressed by the compression unit **102** is discharged to the refrigerant circuit **103**. The compression unit **102** may be

of any type, such as a scroll type, a piston type, a vane type, or the like.

Electric Motor

[0023] The electric motor **M1** includes a motor unit **11**. The motor unit **11** includes a rotor **12** and a stator **13** around which three-phase coils U, V, and W are wound. The motor unit **11** is a three-phase motor including three coils U, V, and W. The motor unit **11** drives the compression unit **102**. The motor unit **11** may be cooled by the refrigerant circulated through the refrigerant circuit **103**.

[0024] The electric motor **M1** includes a motor driving device **10**. The motor driving device **10** includes a battery **BA**, a smoothing capacitor **C**, an inverter **21**, a phase current detector **22**, an input voltage detector **23**, and a controller **30**.

[0025] The inverter **21** includes six switching elements **Q1** to **Q6**, diodes **D1** to **D6**, and a driver **51**. The switching elements **Q1** to **Q6** are, for example, insulated gate bipolar transistors (IGBTs). When the switching elements **Q1** to **Q6** are metal-oxide-semiconductor field-effect transistors (MOSFETs), the switching elements **Q1** to **Q6** are integrated with the diodes **D1** to **D6**. The switching elements **Q1** and **Q2** are connected in series to each other. The switching elements **Q3** and **Q4** are connected in series to each other. The switching elements **Q5** and **Q6** are connected in series to each other. The switching elements **Q1** to **Q6** are connected in parallel to the diodes **D1** to **D6**, respectively. The switching elements **Q1** to **Q6** are each connected to the battery **BA** via the smoothing capacitor **C**.

[0026] A connection line connecting the switching elements **Q1** and **Q2** branches off and is connected to the coil U. A connection line connecting the switching elements **Q3** and **Q4** branches off and is connected to the coil V. A connection line connecting the switching elements **Q5** and **Q6** branches off and is connected to the coil W.

[0027] The driver **51** operates the switching elements **Q1** to **Q6**. The operation of the switching elements **Q1** to **Q6** drives the motor unit **11**. The operation of the switching elements **Q1** to **Q6** refers to switching operations of the switching elements **Q1** to **Q6**, that is, switching of the switching elements **Q1** to **Q6** between on and off states.

[0028] The battery **BA** is a power storage device that can be charged and discharged. The rated voltage of the battery **BA** is, for example, 800V.

[0029] The phase current detector **22** detects a phase current flowing through the motor unit **11**. The phase current detector **22** detects phase currents of at least two phases. In the present embodiment, the phase current detector **22** detects a u-phase current I_u , a v-phase current I_v , and a w-phase current I_w . In another embodiment, the phase current detector **22** may detect phase currents of two phases and calculate a phase current of the remaining phase from the two detected phase currents, taking consideration that the sum of phase currents of the three phases equals zero. The u-phase current I_u , the v-phase current I_v , and the w-phase current I_w are actual currents flowing through the coils U, V, and W of the motor unit **11**.

[0030] The input voltage detector **23** detects an input voltage V_i applied to the inverter **21** by the battery **BA**.

Controller

[0031] The controller **30** includes a processor and memory **80**. The processor is, for example, a central processing unit (CPU), a graphics processing unit (GPU), or a digital signal processor (DSP). The memory **80** includes a random-access memory (RAM) and a read-only memory (ROM). The memory **80** stores program codes or instructions configured to cause the processor to execute processes. The memory **80**, which is a computer-readable medium, includes any medium accessible by a general-purpose computer or a special-purpose computer. The controller **30** may include a hardware circuit, such as an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or the like. The controller **30**, which is processing circuitry, may include one or more processors that operate according to computer programs, one or more hardware circuits (e.g., ASIC or FPGA), or a combination of the above.

[0032] The controller **30** calculates instruction values. The controller **30** uses the instruction values

to control the switching elements Q1 to Q6. The controller 30 drives the motor unit 11 by controlling the inverter 21. In the present embodiment, the controller 30 performs sensorless control that drives the motor unit 11 without using a hardware position sensor that detects a position Hm of the rotor 12 of the motor unit 11. The controller 30 controls the inverter 21 while switching between position estimation using an induced voltage method and position estimation using a harmonic superposition method. The induced voltage method estimates the position Hm of the rotor 12 based on an induced voltage generated in the three-phase coils U, V, and W. The harmonic superposition method estimates the position Hm of the rotor 12 based on an actual current (response current) flowing through the motor unit 11 when a harmonic is superposed on an instruction value. Such control on the inverter 21 drives the motor unit 11. Position estimation by the induced voltage method and position estimation by the harmonic superposition method will now be described. Details of a process executed by the controller 30 to switch the two methods will be described later.

Position Estimation by Induced Voltage Method

[0033] The functionality of the controller 30 when performing position estimation by the induced voltage method is described below. The controller 30 includes a current coordinate converter 31, a position estimator 32, subtractors 33, 35, and 36, a speed controller 34, a current controller 37, and a PWM controller 38.

[0034] The current coordinate converter 31 converts phase currents I_u , I_v , and I_w to a d-axis current I_d and a q-axis current I_q based on the position Hm of the rotor 12 estimated by the position estimator 32. For example, the current coordinate converter 31 converts the phase currents I_u , I_v , and I_w of a three-phase (U, V, and W) fixed coordinate system into currents I_α and I_β of a two-phase (α and β) fixed coordinate system. The current coordinate converter 31 uses the position Hm to convert the currents I_α and I_β into the d-axis current I_d and the q-axis current I_q of a two-phase (d and q) rotating coordinate system. The d-axis and the q-axis are coordinate axes of a d-q coordinate system that rotates together with the rotor 12 of the motor unit 11. The current coordinate converter 31 may directly convert the phase currents I_u , I_v , and I_w to the d-axis current I_d and the q-axis current I_q , without converting them to the currents I_α and I_β .

[0035] The position estimator 32 estimates the position Hm of the rotor 12 of the motor unit 11 based on the d-axis current I_d and the q-axis current I_q output from the current coordinate converter 31, and a d-axis voltage instruction value V_d and a q-axis voltage instruction value V_q output from the current controller 37. For example, the position estimator 32 calculates an induced voltage generated in the coils U, V, and W based on the d-axis current I_d and the q-axis current I_q , the d-axis voltage instruction value V_d and the q-axis voltage instruction value V_q obtained from the current controller 37, and a constant determined by the motor unit 11. Then, the position estimator 32 estimates the position Hm based on the induced voltage. Further, the position estimator 32 estimates a rotation speed F_m of the rotor 12 based on the estimated position Hm. To facilitate understanding, arrows indicating the d-axis current I_d , the d-axis voltage instruction value V_d , and the q-axis voltage instruction value V_q input to the position estimator 32 are not shown in FIG. 1.

[0036] The position estimator 32 uses the induced voltage method to estimate the position of the rotor 12 while the estimated rotation speed F_m is greater than or equal to a predetermined rotation speed F_{mth} . The position estimator 32 uses the harmonic superposition method to estimate the position of the rotor 12 while the estimated rotation speed F_m is less than the predetermined rotation speed F_{mth} . For example, the predetermined rotation speed F_{mth} is a value indicating a rotation speed of the rotor 12 that allows for position estimation of rotor 12 by the induced voltage method. The predetermined rotation speed is, for example, approximately 700 rpm.

[0037] The subtractor 33 calculates a difference ΔF_m between the rotation speed F_m estimated by the position estimator 32 and a rotation speed instruction value F_{mRef} . The rotation speed instruction value F_{mRef} is an instruction value of the rotation speed of the rotor 12 input from outside. For example, the controller 30 receives the rotation speed instruction value F_{mRef} from a

higher-level vehicle control device.

[0038] The speed controller **34** calculates a d-axis current instruction value I_{dRef} and a q-axis current instruction value I_{qRef} based on the difference ΔF_m . For example, the speed controller **34** performs feedback control to calculate the d-axis current instruction value I_{dRef} and the q-axis current instruction value I_{qRef} so that the difference ΔF_m converges to zero. The feedback control is, for example, proportional-integral control.

[0039] The subtractor **35** calculates a difference ΔI_d between the d-axis current instruction value I_{dRef} and the d-axis current I_d . The subtractor **36** calculates a difference ΔI_q between the q-axis current instruction value I_{qRef} and the q-axis current I_q .

[0040] The current controller **37** calculates the d-axis voltage instruction value V_d based on the difference ΔI_d . The current controller **37** calculates the q-axis voltage instruction value V_q based on the difference ΔI_q . For example, the current controller **37** performs feedback control to calculate the d-axis voltage instruction value V_d and the q-axis voltage instruction value V_q so that the difference ΔI_d and the difference ΔI_q converge to zero. The feedback control is, for example, proportional-integral control.

[0041] The PWM controller **38** converts the d-axis voltage instruction value V_d and the q-axis voltage instruction value V_q to a u-phase voltage instruction value V_u , a v-phase voltage instruction value V_v , and a w-phase voltage instruction value V_w based on the position H_m of the rotor **12** estimated by the position estimator **32** and the input voltage V_i . For example, the PWM controller **38** converts the d-axis voltage instruction value V_d and the q-axis voltage instruction value V_q of the d-q coordinate system to voltage instruction values V_α and V_β of a α - β coordinate system. The PWM controller **38** converts the two-phase voltage instruction values V_α and V_β to three-phase voltage instruction values V_u , V_v , and V_w . The PWM controller **38** may directly convert the d-axis voltage instruction value V_d and the q-axis voltage instruction value V_q to the voltage instruction values V_u , V_v , and V_w , without converting them to the voltage instruction values V_α and V_β .

[0042] The inverter **21** is controlled based on the voltage instruction values V_u , V_v , and V_w . Specifically, the PWM controller **38** calculates a duty instruction value based on the voltage instruction values V_u , V_v , and V_w . Subsequently, the PWM controller **38** compares the duty instruction value with a carrier (carrier wave), such as a triangular wave, a sawtooth wave, or the like, to generate a PWM signal having a duty ratio that indicates the duty instruction value. The PWM signal is a gate-on-off signal that determines the ON-OFF states of the switching elements **Q1** to **Q6**. The controller **30** controls the switching elements **Q1** to **Q6** by outputting the PWM signal to corresponding switching elements **Q1** to **Q6** via the driver **51**.

Position Estimation by Harmonic Superposition Method

[0043] The functionality of the controller **30** when performing position estimation by the harmonic superposition method will now be described. As shown in FIG. **1**, in addition to the configuration described above, the controller **30** further includes a harmonic superposer **41**, an adder **42**, a band-stop filter **43**, and a harmonic superposition frequency switcher **44**. The current coordinate converter **31**, the subtractors **33**, **35**, **36**, the speed controller **34**, the current controller **37**, and the PWM controller **38** have the same functionalities as in the induced voltage method. Thus, these components will not be described.

[0044] The harmonic superposer **41** generates a harmonic V_h . The harmonic V_h is defined by $V_a \cdot \cos 2\pi f t$. The “ V_a ” represents the amplitude of the harmonic V_h , and the “ f ” represents the frequency of the harmonic V_h . For example, the frequency f is not dependent on the rotation of the motor unit **11**.

[0045] The adder **42** adds the harmonic V_h generated by the harmonic superposer **41** to the d-axis voltage instruction value V_d . This superposes the harmonic V_h on the d-axis voltage instruction value V_d . In the present embodiment, the controller **30** superposes the harmonic V_h on only the d-axis voltage instruction value V_d . When the position estimator **32** estimates the position of the rotor

12 by the induced voltage method, the harmonic superposer **41** does not superpose the harmonic V_h on the d-axis voltage instruction value V_d . In another embodiment, the controller **30** may estimate the position of the rotor **12** by superposing the harmonic V_h on both the d-axis voltage instruction value V_d and the q-axis voltage instruction value V_q , both the d-axis current instruction value I_{dRef} and the q-axis current instruction value I_{qRef} , or a predetermined instruction value specified by three-phase voltage instruction values V_u , V_v , and V_w .

[0046] The band-stop filter **43** removes a frequency component of a specific band from the d-axis current I_d and the q-axis current I_q . When the harmonic superposer **41** superposes the harmonic V_h on the d-axis voltage instruction value V_d , the d-axis current I_d and the q-axis current I_q contain frequency components based on the harmonic V_h . The band-stop filter **43** removes such frequency components.

[0047] As described above, the position estimator **32** estimates the position of the rotor **12** using the harmonic superposition method while the estimated rotation speed F_m is less than the predetermined rotation speed F_{mth} . Specifically, the position estimator **32** estimates the position H_m of the rotor **12** and the rotation speed F_m of the rotor **12** from the q-axis current I_q calculated by the current coordinate converter **31**. When the harmonic V_h is superposed on the d-axis voltage instruction value V_d , the q-axis current I_q contains a current harmonic based on the harmonic V_h . The position estimator **32** estimates the position H_m of the rotor **12** and the rotation speed F_m of the rotor **12** based on the current harmonic contained in the q-axis current I_q and a mathematical expression model of the motor unit **11**. For example, the position estimator **32** calculates an axis error $\Delta\theta_c$ from the current harmonic. The axis error $\Delta\theta_c$ is a difference between the actual position H_m of the rotor **12** and the position H_m of the rotor **12** recognized by the controller **30**. The position estimator **32** estimates the position H_m of the rotor **12** and the rotation speed F_m of the rotor **12** so that the axis error $\Delta\theta_c$ becomes equal to zero.

[0048] The harmonic superposition frequency switcher **44** switches the frequency of the harmonic V_h based on the phase currents I_u , I_v , and I_w detected by the phase current detector **22** and a first threshold value $Th1$.

Relationship between Frequency of Harmonic V_h and First Threshold Value $Th1$

[0049] FIG. 2 is a graph illustrating the relationship between the frequency of the harmonic V_h and an effective value of the phase current of the motor unit **11**. The frequency of the harmonic V_h includes a first frequency $f1$ and a second frequency $f2$. The second frequency $f2$ is lower than the first frequency $f1$. In the description hereafter, the first frequency $f1$ may be referred to as the initial superposition frequency.

[0050] As shown in FIG. 2, while each of the phase currents I_u , I_v , and I_w of the motor unit **11** is in a range of zero to the first threshold value $Th1$, the harmonic superposition frequency switcher **44** causes the harmonic superposer **41** to generate the harmonic V_h having the first frequency $f1$ so that the harmonic V_h having the first frequency $f1$ is superposed on the d-axis voltage instruction value V_d . When any of the phase currents I_u , I_v , and I_w detected by the phase current detector **22** becomes greater than the first threshold value $Th1$, the harmonic superposition frequency switcher **44** switches the frequency of the harmonic V_h , which is to be generated by the harmonic superposer **41**, from the first frequency $f1$ to the second frequency $f2$. As a result, the harmonic V_h having the second frequency $f2$ is superposed on the d-axis voltage instruction value V_d . The first threshold value $Th1$ is based on the heat resistance temperature of the motor unit **11**. The first threshold value $Th1$ indicates a phase current value when the temperature of the motor unit **11** is higher than the average temperature of the motor unit **11** in a normal state by a predetermined value.

[0051] As the phase currents I_u , I_v , and I_w of the motor unit **11** increase, the temperature of the motor unit **11** increases. That is, the phase currents I_u , I_v , and I_w of the motor unit **11** correlate with the temperature of the motor unit **11**. Therefore, the phase current detector **22** is an example of a temperature-associated value obtainer, and a detection result of the phase currents I_u , I_v , and I_w is

an example of a temperature-associated value that is associated with the temperature of the motor unit **11**. In other words, a process that determines whether any of the phase currents I_u , I_v , and I_w detected by the phase current detector **22** becomes greater than the first threshold value $Th1$ is an example of a process that determines whether the temperature of the motor unit **11** becomes greater than the first threshold value $Th1$.

Process of Controller **30**

[0052] The series of processes executed by the controller **30** to control the motor unit **11** will now be described with reference to FIG. **3**. The process of the flowchart shown in FIG. **3** is executed in predetermined calculation cycles from when the motor unit **11** is started to when the motor unit **11** is stopped. For example, the motor unit **11** is started when the vehicle air conditioner **100** is started, and is stopped when the vehicle air conditioner **100** is stopped.

[0053] In the series of processes related to the control of the motor unit **11**, the controller **30** first obtains the rotation speed F_m of the rotor **12** (step **S1**). In the initial calculation cycle after the motor unit **11** is started, the controller **30** may estimate the rotation speed F_m using the harmonic superposition method or the induced voltage method.

[0054] Next, the controller **30** determines whether the rotation speed F_m is greater than or equal to the predetermined rotation speed F_{mth} (step **S2**). When the rotation speed F_m is greater than or equal to the predetermined rotation speed F_{mth} (step **S2**: YES), the controller **30** uses the induced voltage method to estimate the rotation position of the rotor **12** (step **S3**). When the rotation speed F_m is less than the predetermined rotation speed F_{mth} (step **S2**: NO), the controller **30** uses the harmonic superposition method to estimate the rotation position of the rotor **12** (step **S4**).

Subsequently, the controller **30** generates a PWM signal based on the rotation speed F_m estimated in step **S3** or step **S4**, and then controls the inverter **21** to drive the motor unit **11** (step **S5**).

[0055] Also, the controller **30** executes the series of processes illustrated in FIG. **4**, simultaneously with the flowchart in FIG. **3**, to set the frequency of the harmonic V_h . The process of the flowchart shown in FIG. **4** may be executed in the same calculation cycles as the flowchart in FIG. **3**, or may be executed in different calculation cycles. The process of the flowchart shown in FIG. **4** does not have to be executed simultaneously with the flowchart in FIG. **3**. For example, the process of the flowchart shown in FIG. **4** may be a subroutine executed within the flowchart in FIG. **3** (for example, executed before step **S4**).

[0056] When the motor unit **11** is started, the frequency of the harmonic V_h is set to the initial superposition frequency (first frequency $f1$). Therefore, if the motor unit **11** was stopped after the frequency of the harmonic V_h had been switched to the second frequency $f2$, the frequency of the harmonic V_h is switched from the second frequency $f2$ to the initial superposition frequency when the motor unit **11** is started again.

[0057] In the series of processes related to the setting of the frequency of the harmonic V_h , the phase current detector **22** first obtains the phase currents I_u , I_v , and I_w (step **S102**). Next, the harmonic superposition frequency switcher **44** determines whether the phase currents I_u , I_v , and I_w are increased based on the detection result of the phase current detector **22** (step **S104**). When none of the phase currents I_u , I_v , and I_w is increased (step **S104**: NO), there is no need to switch the frequency of the harmonic V_h . Thus, the harmonic superposition frequency switcher **44** ends the series of processes.

[0058] When the harmonic superposition frequency switcher **44** determines that any of the phase currents I_u , I_v , and I_w is increased (step **S104**: YES), the harmonic superposition frequency switcher **44** determines whether any of the phase currents I_u , I_v , and I_w has become greater than the first threshold value $Th1$ (step **S106**). When the harmonic superposition frequency switcher **44** determines that none of the phase currents I_u , I_v , and I_w has become greater than the first threshold value $Th1$ (step **S106**: NO), the harmonic superposition frequency switcher **44** ends the series of processes. When the harmonic superposition frequency switcher **44** determines that any of the phase currents I_u , I_v , and I_w has become greater than the first threshold value $Th1$ (step **S106**:

YES), the harmonic superposition frequency switcher **44** switches the frequency of the harmonic V_h from the first frequency f_1 to the second frequency f_2 (step **S108**). In this manner, the harmonic superposer **41** superposes the harmonic V_h having the second frequency f_2 on the d-axis voltage instruction value V_d . Then, the controller **30** ends the series of processes.

Advantages of the Embodiment

[0059] The above embodiment has the following advantages. [0060] (1) The position estimator **32** estimates the position of the rotor **12** by the harmonic superposition method. The frequency of the harmonic V_h is set to the first frequency f_1 when the motor unit **11** is started. When any of the phase currents I_u , I_v , and I_w exceeds the first threshold value Th_1 , the harmonic superposition frequency switcher **44** determines that the temperature of the motor unit **11** has become greater than the first threshold value. Then, the harmonic superposition frequency switcher **44** switches the frequency of the harmonic V_h , which is to be superposed on the d-axis voltage instruction value V_d , from the first frequency f_1 to the second frequency f_2 .

[0061] With this configuration, the frequency of the harmonic V_h is set relatively high when position estimation of the rotor **12** by the harmonic superposition method is started. This minimizes the adverse effects on the NV characteristics. Furthermore, the frequency of the harmonic V_h is lowered when the temperature of the motor unit **11** increases. This avoids overheating of the motor unit **11**. [0062] (2) The detection result of the phase current detector **22**, which detects the phase currents I_u , I_v , and I_w flowing through the motor unit **11**, is used as the temperature-associated value that is associated with the temperature of the motor unit **11**.

[0063] This configuration obtains the temperature-associated value of the motor unit **11** without an additional configuration for detecting the temperature of the motor unit **11**. [0064] (3) The position estimator **32** estimates the position of the rotor **12** using the harmonic superposition method or the induced voltage method. Specifically, when the rotation speed F_m of the rotor **12** based on the estimation result is greater than or equal to the predetermined rotation speed F_{mth} , the position estimator **32** uses the induced voltage method to estimate the position of the rotor **12**. When the rotation speed F_m of the rotor **12** based on the estimation result is less than the predetermined rotation speed F_{mth} , the position estimator **32** uses the harmonic superposition method to estimate the position of the rotor **12**. When the position estimator **32** estimates the position of the rotor **12** by the induced voltage method, the harmonic superposer **41** does not superpose the harmonic V_h on the d-axis voltage instruction value V_d .

[0065] As the rotation speed F_m of the rotor **12** decreases, the induced voltage decreases.

Therefore, if the rotation speed F_m of the rotor **12** is relatively low when the position estimator **32** estimates the position of the rotor **12** using the induced voltage method, the estimation accuracy may become lower. In this respect, the above configuration allows the position estimator **32** to accurately estimate the position of the rotor **12** using an appropriate one of the harmonic superposition method and the induced voltage method in accordance with the rotation speed F_m of the rotor **12**. Furthermore, if the rotational speed F_m of the rotor **12** is relatively low when the refrigerant for cooling the motor unit **11** is circulated by the rotation of the motor unit **11**, the flow rate of refrigerant may be reduced. This may cause overheating of the motor unit **11**. Therefore, it is highly advantageous to lower the frequency of the harmonic V_h when the temperature of the motor unit **11** is relatively high, as in the above embodiment. [0066] (4) The motor-driven compressor **101** includes the compression unit **102** that compresses the refrigerant, which is a fluid, and the electric motor **M1** that drives the compression unit **102**. The frequency of the harmonic V_h is switched when estimating the position of the rotor **12** by the harmonic superposition method, as described above. This avoids overheating of the motor unit **11** and minimizes the adverse effects on the NV characteristics. With this configuration, the vehicle air conditioner **100** including the motor-driven compressor **101** avoids overheating of the motor unit **11** and reduces the noise and vibration experienced by occupants in the passenger compartment.

[0067] The above embodiment may be modified as follows. The above embodiment and the

following modifications can be combined as long as the combined modifications remain technically consistent with each other.

[0068] The harmonic superposition frequency switcher **44** switches the frequency of the harmonic V_h from the first frequency f_1 to the second frequency f_2 when the phase currents I_u , I_v , and I_w of the motor unit **11** are increased. However, there is no limit to such a configuration. The harmonic superposition frequency switcher **44** may switch the frequency of the harmonic V_h from the second frequency f_2 to the first frequency f_1 when the phase currents I_u , I_v , and I_w of the motor unit **11** are decreased. The details will now be described with reference to FIGS. 5 to 6.

[0069] When any of the phase currents I_u , I_v , and I_w of the motor unit **11** becomes less than or equal to the first threshold value Th_1 , the harmonic superposition frequency switcher **44** maintains the frequency of the harmonic V_h at the second frequency f_2 while the phase currents I_u , I_v , and I_w are greater than or equal to a second threshold value Th_2 . The second threshold value Th_2 is less than the first threshold value Th_1 . When the frequency of the harmonic V_h is the second frequency f_2 , the harmonic superposition frequency switcher **44** switches the frequency of the harmonic V_h to the first frequency f_1 after any of the phase currents I_u , I_v , and I_w of the motor unit **11** becomes less than the second threshold value Th_2 . As described above, if any of the phase currents I_u , I_v , and I_w of the motor unit **11** becomes greater than the first threshold value Th_1 when the frequency of the harmonic V_h is the first frequency f_1 , the harmonic superposition frequency switcher **44** switches the frequency of the harmonic V_h to the second frequency f_2 . That is, hysteresis is provided in the switching of the frequency of the harmonic V_h . A process that determines whether any of the phase currents I_u , I_v , and I_w detected by the phase current detector **22** becomes less than the second threshold value Th_2 is an example of a process that determines whether the temperature-associated value of the motor unit **11** becomes less than the second threshold value Th_2 .

[0070] The series of processes related to the setting of the frequency of the harmonic V_h in accordance with the modified example will now be described with reference to FIG. 6. The flowchart shown in FIG. 6 includes steps **S200** to **S202**, in addition to the process of the flowchart in FIG. 4.

[0071] When the phase currents I_u , I_v , and I_w are not increased (step **S104**: NO), the harmonic superposition frequency switcher **44** determines that the phase currents I_u , I_v , and I_w are decreased and determines whether any of the phase currents I_u , I_v , and I_w is less than the second threshold value Th_2 (step **S200**). When the harmonic superposition frequency switcher **44** determines that none of the phase currents I_u , I_v , and I_w is less than the second threshold value Th_2 (step **S200**: NO), the harmonic superposition frequency switcher **44** ends the series of processes. When the harmonic superposition frequency switcher **44** determines that any of the phase currents I_u , I_v , and I_w is less than the second threshold value Th_2 (step **S200**: YES), the harmonic superposition frequency switcher **44** switches the frequency of the harmonic V_h from the second frequency f_2 to the first frequency f_1 (step **S202**). In this manner, the harmonic superposer **41** superposes the harmonic V_h having the first frequency f_1 on the d-axis voltage instruction value V_d . Then, the controller **30** ends the series of processes.

[0072] This configuration appropriately switches the frequency of the harmonic V_h in accordance with the temperature of the motor unit **11**. Further, hysteresis is provided in the switching of the frequency of the harmonic V_h , thereby avoiding frequent switching of the frequency of the harmonic V_h .

[0073] In the present modified example, the frequency of the harmonic V_h may be set to the second frequency f_2 when the motor unit **11** is started.

[0074] The detection result of the phase currents I_u , I_v , and I_w detected by the phase current detector **22** is used as the temperature-associated value of the motor unit **11**. However, there is no limit to such a configuration. Instead of (or in addition to) the detection result of the phase currents I_u , I_v , and I_w , other types of information may be used as the temperature-associated value of the

motor unit **11**. For example, in addition to the above configuration, the motor driving device **10** may include a temperature detector attached to the motor unit **11** and configured to detect the temperature of the motor unit **11**. The temperature detector is implemented by, for example, a thermocouple and a functional unit that performs AD conversion on a measurement result of the thermocouple.

[0075] In this case, a detection result of the temperature detector is used as the temperature-associated value of the motor unit **11**. The harmonic superposition frequency switcher **44** compares the temperature-associated value of the motor unit **11**, which is based on the detection result of the temperature detector, with the first threshold value Th1 and the second threshold value Th2 to switch the frequency of the harmonic Vh. In this case, the first threshold value Th1 and the second threshold value Th2 are based on the heat resistance temperature of the motor unit **11**. Specifically, the first threshold value Th1 indicates a temperature higher than the average temperature of the motor unit **11** in a normal state. The second threshold value Th2 is less than the first threshold value Th1.

[0076] The first threshold value Th1 does not have to be based on the average temperature of the motor unit **11** in a normal state. The first threshold value Th1 may be based on a statistical value, such as the median value, average value, mode value, or the like, of a temperature range achievable by the motor unit **11**. Alternatively, the first threshold value Th1 may be based on a statistical value, such as the median value, average value, or the like, of a heat resistance temperature range of the motor unit **11**.

[0077] The harmonic superposition frequency switcher **44** switches the frequency of the harmonic Vh based on the effective value of any of the phase currents Iu, Iv, and Iw detected by the phase current detector **22**. However, there is no limit to such a configuration. The harmonic superposition frequency switcher **44** may switch the frequency of the harmonic Vh based on a statistical value, such as the median value, average value, maximum value, minimum value, or the like, of the phase currents Iu, Iv, and Iw detected by the phase current detector **22**.

[0078] The position estimator **32** estimates the position of the rotor **12** using the harmonic superposition method or the induced voltage method. However, there is no limit to such a configuration. The position estimator **32** may estimate the position of the rotor **12** using only the harmonic superposition method.

[0079] The harmonic superposer **41** superposes the harmonic Vh on the d-axis voltage instruction value Vd. However, there is no limit to such a configuration. The harmonic superposer **41** may superpose the harmonic Vh on the current instruction values IdRef and IqRef.

[0080] Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

Claims

1. An electric motor comprising: a motor unit including a rotor and a stator around which three-phase coils are wound; an inverter including a driver and a switching element operated by the driver, the inverter being configured to drive the motor unit through operation of the switching element; a temperature-associated value obtainer configured to obtain a temperature-associated value that is associated with a temperature of the motor unit; and processing circuitry configured to

calculate instruction values for control of the switching element and control the switching element, wherein the processing circuitry is configured to: superpose a harmonic on at least one of the instruction values; and estimate a position of the rotor by a harmonic superposition method based on the harmonic superposed on the at least one of instruction values, a frequency of the harmonic includes a first frequency and a second frequency that is lower than the first frequency, and the processing circuitry is further configured to: determine whether the temperature-associated value becomes greater than a threshold value; and set the frequency of the harmonic for superposition on the at least one of the instruction values to the second frequency when the temperature-associated value becomes greater than the threshold value.

2. The electric motor according to claim 1, wherein the frequency of the harmonic is set to the first frequency when the motor unit is started.

3. The electric motor according to claim 1, wherein the threshold value is a first threshold value, and a second threshold value is less than the first threshold value, and the processing circuitry is further configured to: determine whether the temperature-associated value becomes less than the second threshold value when the frequency of the harmonic is the second frequency; and set the frequency of the harmonic to the first frequency when the temperature-associated value becomes less than the second threshold value.

4. The electric motor according to claim 1, wherein the temperature-associated value obtainer includes a phase current detector configured to detect a phase current flowing through the motor unit, and the temperature-associated value includes a detection result of the phase current detector.

5. The electric motor according to claim 1, wherein the temperature-associated value obtainer includes a temperature detector attached to the motor unit and configured to detect the temperature of the motor unit, and the temperature-associated value includes a detection result of the temperature detector.

6. The electric motor according to claim 1, wherein the processing circuitry is further configured to: estimate the position of the rotor by an induced voltage method based on an induced voltage generated in the three-phase coils when a rotation speed of the rotor is greater than or equal to a predetermined rotation speed; and estimate the position of the rotor by the harmonic superposition method when the rotation speed is less than the predetermined rotation speed, and when the position of the rotor is estimated by the induced voltage method, the processing circuitry is further configured to not perform superposition of the harmonic.

7. A motor-driven compressor, comprising: a compression unit configured to compress a fluid; and the electric motor according to claim 1 and configured to drive the compression unit.
